Supplementary material

Establishing the content in trace and minor elements of magnetite as a biosignature of magnetotactic bacteria

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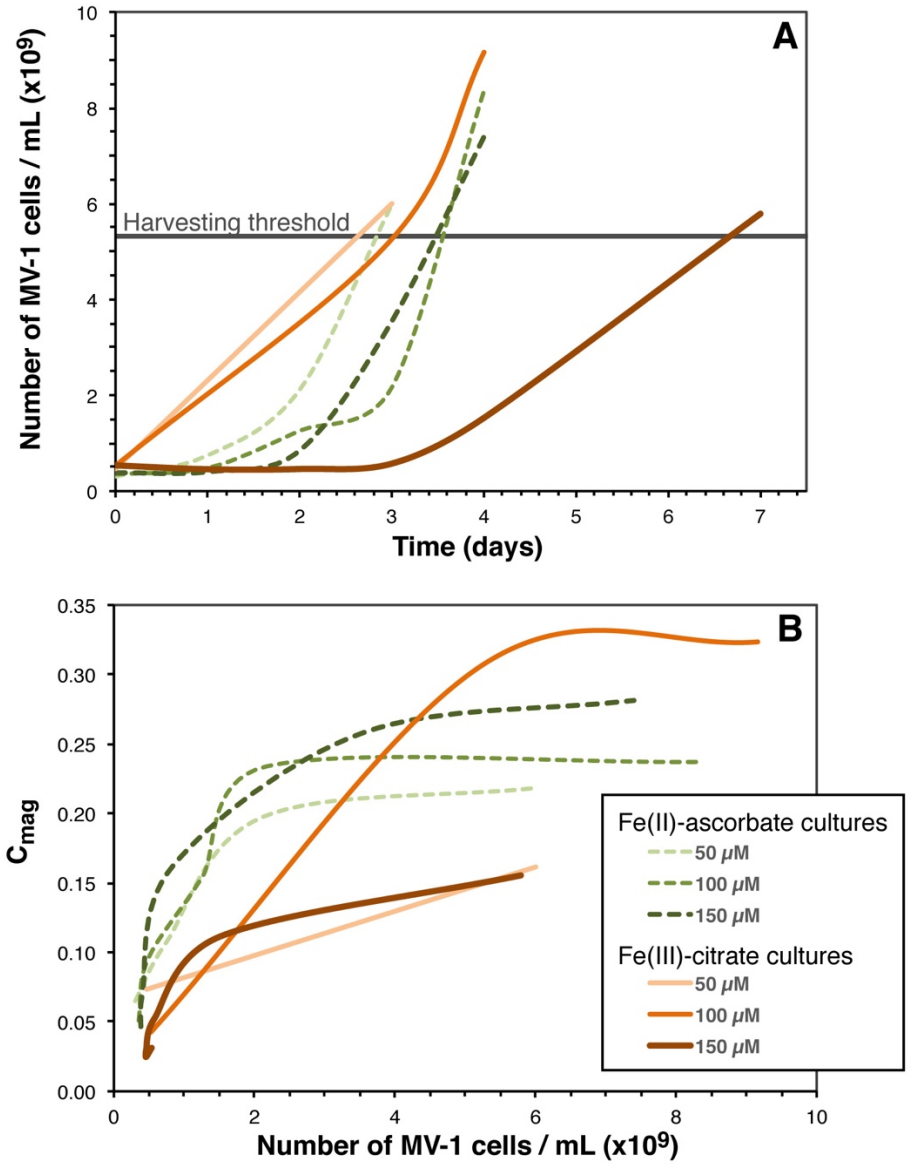


Fig. S1. Monitoring of MV1 cultures for various concentrations of initial Fe(II)-ascorbate (green dotted lines) and Fe(IIII)-citrate (orange lines) conditions. (A) Evolution of the total number of bacteria per milliliter, calculated from OD measurements (OD = 560 nm). (B) Proportion of magnetic cells during bacterial growth (estimated from C_{mag} measurements, see main text for detailed description). When bacterial concentration increases by more than four times relative to initial condition, the proportion of magnetic cells over the whole population becomes relatively stable. In cultures with Fe(II)-ascorbate, the proportion of magnetic cells is correlated with initial iron concentration, suggesting a better efficiency in producing magnetosome at higher Fe concentration.

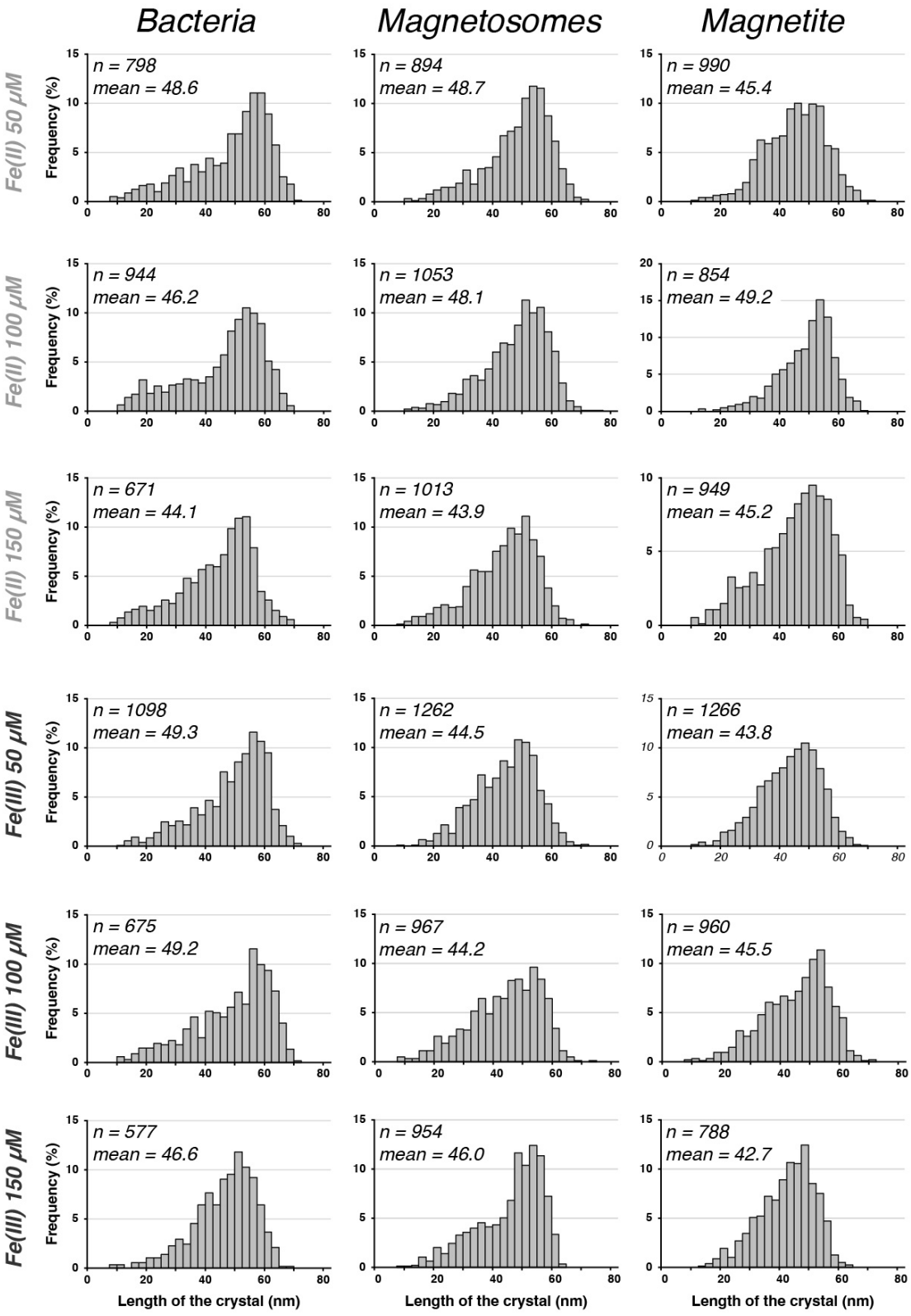


Fig. S2. Size distribution of magnetite nanocrystals produced by MV-1 inside bacterial cells before harvesting, after purification treatment in magnetosome samples, and after detergent treatment to remove membranes of magnetosomes (*i.e.* purified magnetite). The results show identical size distribution at all steps, illustrating a limited bias from analytical treatments.

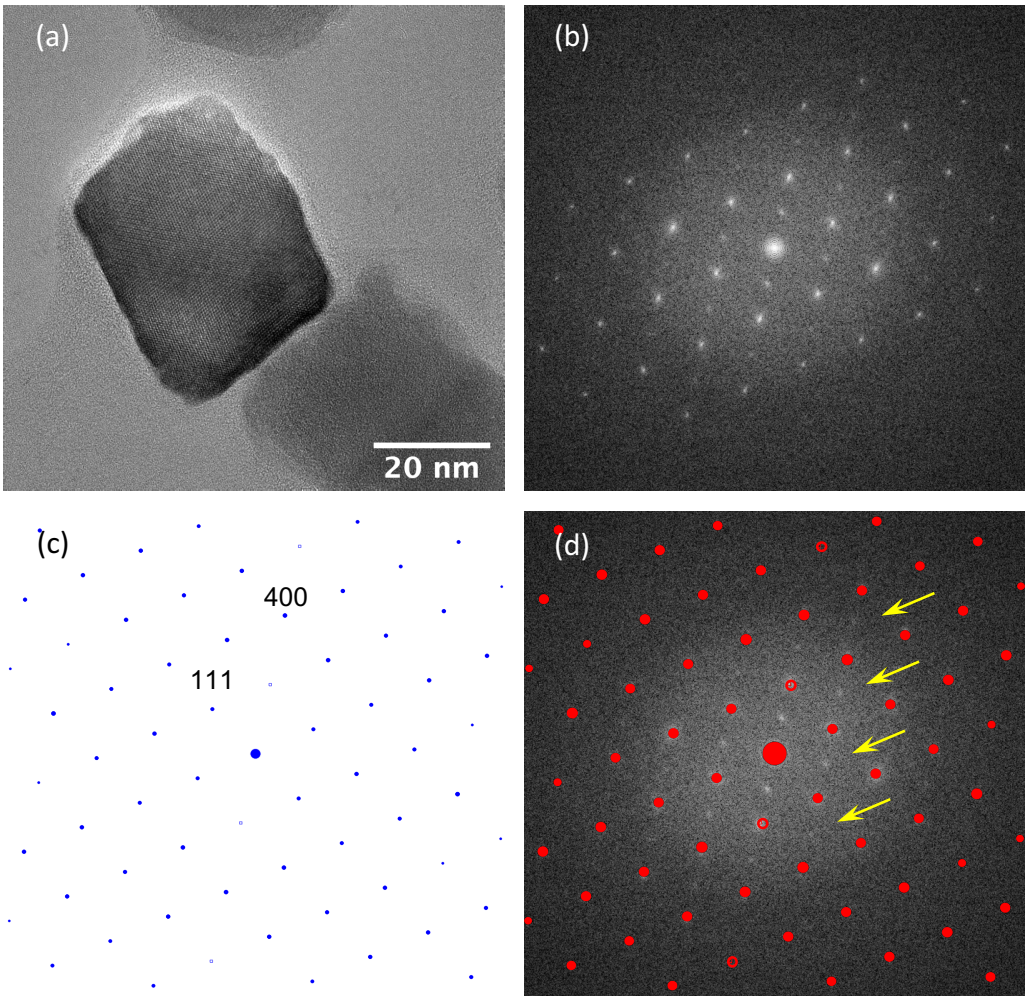


Fig. S3. (a) HR-TEM image of a magnetite nanocrystal produced from MV-1 culture with Fe(II)-ascorbate condition at 50 μM. (b) Fourier transform (FFT) corresponding to the central zone of magnetosome. (c) Simulated diffraction pattern for magnetite along the crystallographic axis <110>. (d) Comparison between FFT-simulation and analyzed pattern showing a few additional points specific to maghemite.

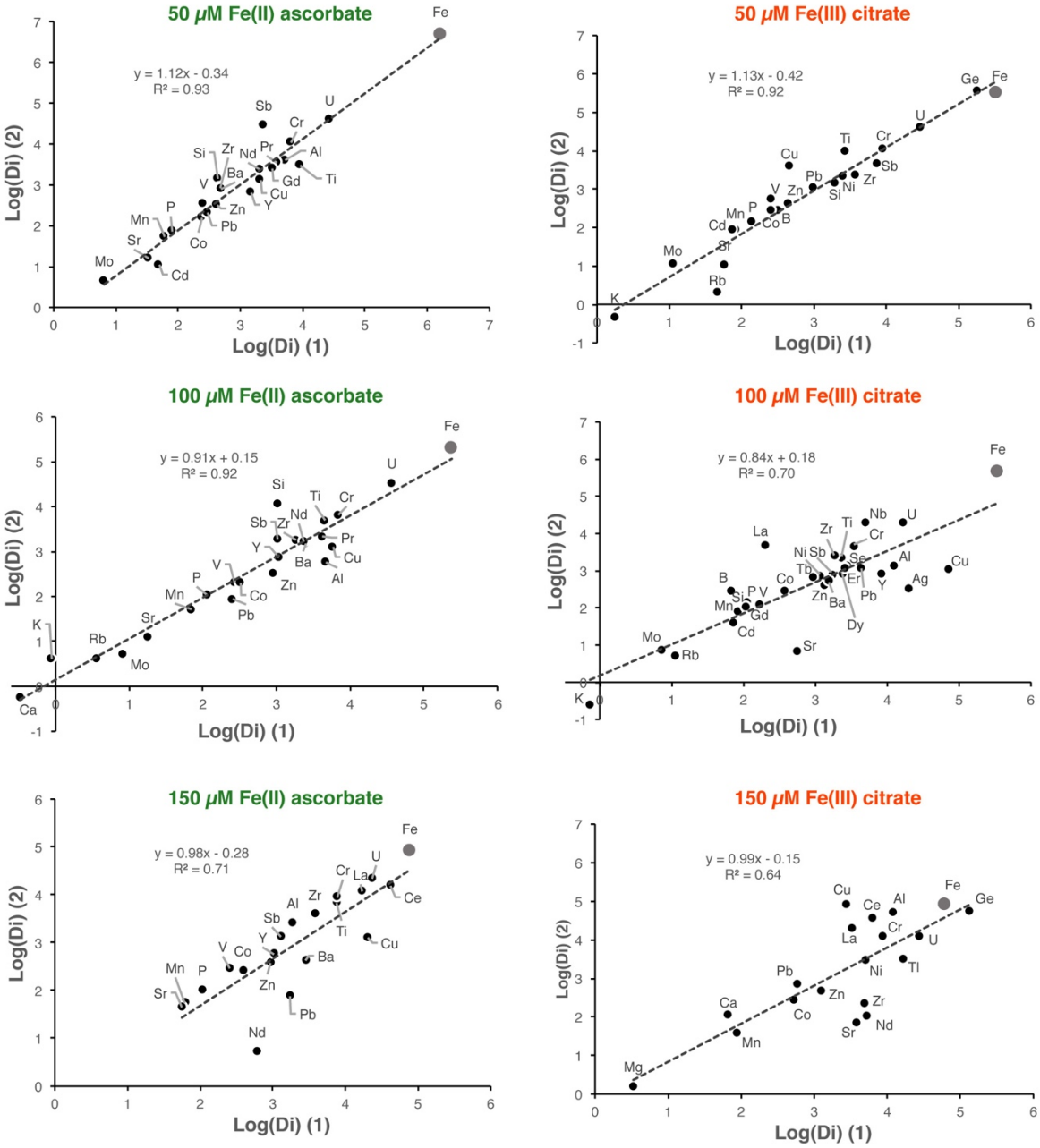


Fig. S4. Comparison of the distribution coefficients (D_i) obtained for replicates (1) and (2) in all culture conditions. Green and orange colors correspond to Fe(II)-ascorbate (left panel) and Fe(III)-citrate (right panel) conditions, respectively. The duplicates show a good correlation with a slope near 1 (dark grey dotted lines and equations on the graph).

Table S1. Growth parameters monitored during Fe(II)-ascorbate and Fe(III)-citrate cultures of MV-1 (cell concentration, pH, Cmag). R#1 to R#10 are individual replicates (1L bottle). Sub-samples were collected once a day for most experiments.

	Day	R#1	R#2	R#3	R#4	R#5	R#6	R#8	R#9	R#10	Average	± 1SD
Fe(II)-ascorbate												
50 uM Fe(II)												
cells/L	0	3,52E+08			2,59E+08			3,06E+08			3,06E+08	4,63E+07
	1	6,45E+08	1,00E+09	7,53E+08	6,30E+08	6,60E+08	8,92E+08	7,69E+08	7,84E+08	5,83E+08	7,46E+08	1,35E+08
	2	1,60E+09	2,94E+09	2,08E+09	1,86E+09	2,02E+09	2,53E+09	2,14E+09	2,11E+09	1,77E+09	2,12E+09	4,05E+08
	3	4,61E+09	7,03E+09	6,22E+09	5,60E+09	6,56E+09	6,71E+09	6,51E+09	5,66E+09	5,00E+09	5,99E+09	8,21E+08
pH	0	6,96	6,97	6,97	6,98	6,95	6,98	6,99	6,95	7,00	6,97	0,02
	1	6,96	6,96	6,97	6,96	6,95	6,96	6,98	6,94	7,00	6,96	0,02
	2	6,95	7,02	7,00	6,97	6,98	7,02	7,00	6,99	7,00	6,99	0,02
	3	7,07	7,05	7,00	6,97	7,02	7,02	7,05	6,99	6,99	7,02	0,03
Cmag	0	0,057			0,078			0,060			0,065	0,011
	1	0,090	0,107	0,104	0,114	0,122	0,105	0,103	0,102	0,118	0,107	0,010
	2	0,179	0,210	0,198	0,202	0,202	0,194	0,210	0,190	0,191	0,197	0,010
	3	0,208	0,211	0,220	0,227	0,216	0,216	0,232	0,213	0,221	0,218	0,008
100 uM Fe(II)												
cells/L	0	3,98E+08	2,90E+08	3,83E+08	3,36E+08	4,14E+08	3,52E+08	4,14E+08	2,28E+08	3,67E+08	3,54E+08	6,15E+07
	1	5,06E+08	4,60E+08	4,91E+08	4,60E+08	5,37E+08	5,99E+08	4,75E+08	5,22E+08	2,90E+08	4,82E+08	8,42E+07
	2	1,26E+09	1,09E+09	1,17E+09	1,14E+09	1,65E+09	1,63E+09	1,32E+09	8,46E+08	1,25E+09	1,26E+09	2,54E+08
	3	1,86E+09	1,93E+09	2,22E+09	2,33E+09	2,20E+09	3,55E+09	2,22E+09	1,37E+09	1,83E+09	2,17E+09	5,95E+08
	4	8,52E+09	8,78E+09	8,62E+09	8,22E+09	9,35E+09	6,60E+09	7,53E+09	8,87E+09	8,76E+09	8,36E+09	8,27E+08
pH	0	7,01	7,00	6,99	6,99	6,98	6,97	6,98	6,99	6,99	6,99	0,01
	1	6,96	6,95	6,95	6,95	6,96	6,96	6,95	6,97	6,96	6,96	0,01
	2	6,89	6,89	6,89	6,88	6,89	6,87	6,90	6,89	6,89	6,89	0,01
	3	6,94	6,92	6,87	6,91	6,90	6,94	6,90	6,89	6,89	6,91	0,02
	4	7,08	7,03	6,97	7,02	7,04	6,97	7,01	7,00	7,03	7,02	0,03
Cmag	0	0,055	0,061	0,056	0,043	0,054	0,057	0,054	0,065	0,014	0,051	0,015
	1	0,088	0,091	0,089	0,091	0,085	0,116	0,090	0,099	0,106	0,095	0,010
	2	0,124	0,161	0,171	0,182	0,110	0,209	0,143	0,157	0,133	0,154	0,031
	3	0,232	0,233	0,246	0,237	0,205	0,271	0,230	0,221	0,229	0,234	0,018
	4	0,229	0,237	0,229	0,240	0,240	0,259	0,234	0,235	0,231	0,237	0,009
150 uM Fe(II)												
cells/L	0	3,98E+08	6,30E+08	3,67E+08	3,98E+08	3,67E+08	3,52E+08	3,06E+08	3,06E+08	3,36E+08	3,84E+08	9,80E+07
	1	2,90E+08	5,52E+08	5,37E+08	3,52E+08	3,83E+08	3,67E+08	3,67E+08	4,44E+08	3,36E+08	4,03E+08	9,00E+07
	2	7,69E+08	7,99E+08	7,53E+08	8,30E+08	9,54E+08	7,99E+08	7,38E+08	1,31E+09	6,76E+08	8,47E+08	1,89E+08
	3	2,61E+09	2,74E+09	2,73E+09	3,73E+09	4,84E+09	3,22E+09	2,70E+09	6,86E+09	2,48E+09	3,55E+09	1,45E+09
	4	6,65E+09	7,61E+09	8,58E+09	7,25E+09	7,00E+09	8,75E+09	6,74E+09	7,19E+09	6,83E+09	7,40E+09	7,74E+08
pH	0	7,07	7,06	7,07	7,09	7,08	7,08	7,07	7,04	7,11	7,07	0,02
	1	6,97	6,97	6,98	6,93	6,95	6,96	6,95	6,95	6,99	6,96	0,02
	2	6,94	6,93	6,97	6,90	6,88	6,89	6,89	6,87	6,94	6,91	0,03
	3	6,94	6,93	6,98	6,95	6,95	6,93	6,94	7,00	6,97	6,95	0,02
	4	7,02	7,00	7,09	7,03	6,97	7,01	7,02	6,97	7,02	7,01	0,04
Cmag	0	0,055	0,034	0,056	0,041	0,042	0,043	0,045	0,045	0,058	0,047	0,008
	1	0,061	0,072	0,073	0,086	0,069	0,056	0,070	0,079	0,058	0,069	0,010
	2	0,134	0,141	0,156	0,178	0,220	0,152	0,116	0,235	0,132	0,163	0,041
	3	0,264	0,213	0,246	0,301	0,302	0,266	0,203	0,293	0,236	0,258	0,037
	4	0,274	0,265	0,267	0,304	0,311	0,279	0,273	0,300	0,259	0,281	0,019

Table S1. Continue.

	Day	R#1	R#2	R#3	R#4	R#5	R#6	R#8	R#9	R#10	Average	± 1SD
Fe(III)-citrate												
50 uM Fe(III)												
cells/L	0	3,67E+08	4,75E+08	3,83E+08	4,60E+08	5,06E+08	3,83E+08	5,22E+08	6,45E+08	4,75E+08	4,68E+08	8,70E+07
	3	6,86E+09	3,52E+09	8,49E+09	4,33E+09	6,20E+09	5,92E+09	7,54E+09	3,92E+09	7,24E+09	6,00E+09	1,74E+09
pH	0	7,01	6,96	7,03	7,00	7,04	7,02	7,01	6,97	7,03	7,01	0,03
	3	6,95	6,90	6,94	6,93	6,97	6,97	7,01	6,94	6,94	6,95	0,03
Cmag	0	0,085	0,077	0,069	0,078	0,075	0,069	0,074	0,056	0,077	0,073	0,008
	3	0,157	0,167	0,146	0,171	0,167	0,160	0,159	0,166	0,161	0,161	0,008
100 uM Fe(III)												
cells/L	0	3,67E+08	5,22E+08	5,37E+08	6,60E+08	6,45E+08	4,29E+08	6,30E+08	5,22E+08	4,44E+08	5,28E+08	1,03E+08
	3	2,77E+09	3,90E+09	7,03E+09	4,58E+09	4,73E+09	4,26E+09	7,90E+09	8,04E+09	4,15E+09	5,26E+09	1,90E+09
	4	7,08E+09	1,05E+10	7,64E+09	9,43E+09	9,23E+09	1,03E+10	7,99E+09	9,06E+09	1,12E+10	9,16E+09	1,39E+09
pH	0	7,01	7,02	6,99	6,99	6,98	7,04	6,99	7,00	7,02	7,00	0,02
	3	6,94	6,97	6,97	6,96	6,96	6,91	6,98	6,97	6,90	6,95	0,03
	4	6,92	6,97	6,94	6,93	6,95	6,99	6,91	6,95	7,00	6,95	0,03
Cmag	0	0,056	0,049	0,037	0,044	0,022	0,040	0,045	0,049	0,039	0,043	0,010
	3	0,317	0,277	0,346	0,343	0,347	0,226	0,349	0,292	0,269	0,307	0,044
	4	0,318	0,278	0,356	0,392	0,389	0,242	0,363	0,292	0,281	0,323	0,054
150 uM Fe(III)												
cells/L	0	5,06E+08	4,75E+08	5,37E+08	5,37E+08	5,37E+08	6,14E+08	4,75E+08	5,99E+08	5,52E+08	5,37E+08	4,82E+07
	1	3,67E+08	3,67E+08	4,91E+08	4,75E+08	4,29E+08	4,91E+08	4,91E+08	5,06E+08	4,91E+08	4,56E+08	5,50E+07
	2	3,52E+08	4,14E+08	4,60E+08	4,75E+08	4,14E+08	5,06E+08	4,75E+08	5,22E+08	5,37E+08	4,62E+08	5,95E+07
	3	3,67E+08	4,91E+08	6,14E+08	6,14E+08	5,52E+08	5,99E+08	5,68E+08	6,60E+08	7,07E+08	5,75E+08	9,95E+07
	4	6,45E+08	1,20E+09	1,83E+09	1,85E+09	1,42E+09	1,42E+09	1,60E+09	1,69E+09	2,11E+09	1,53E+09	4,30E+08
	7	8,33E+09	7,20E+09	4,90E+09	4,15E+09	5,28E+09	3,96E+09	7,44E+09	3,27E+09	7,57E+09	5,79E+09	1,86E+09
pH	0	6,95	6,95	6,91	6,93	6,94	6,93	6,96	6,92	6,96	6,94	0,02
	1	6,98	6,97	6,93	6,95	6,96	6,95	6,96	6,95	6,97	6,96	0,01
	2	6,97	6,95	6,92	6,94	6,95	6,95	6,96	6,95	6,96	6,95	0,01
	3	6,97	6,93	6,90	6,93	6,94	6,93	6,94	6,92	6,92	6,93	0,02
	4	6,93	6,91	6,87	6,91	6,90	6,90	6,89	6,91	6,92	6,90	0,02
	7	6,99	7,00	6,96	6,88	6,93	6,92	6,97	6,90	6,97	6,95	0,04
Cmag	0	0,038	0,026	0,037	0,024	0,037	0,023	0,038	0,035	0,024		0,007
	1	0,028	0,028	0,025	0,026	0,027	0,025	0,013	0,025	0,025	0,025	0,005
	2	0,029	0,027	0,026	0,026	0,041	0,025	0,038	0,025	0,037	0,030	0,006
	3	0,028	0,063	0,057	0,057	0,048	0,047	0,060	0,044	0,065	0,052	0,012
	4	0,090	0,120	0,120	0,132	0,108	0,101	0,113	0,108	0,114	0,112	0,012
	7	0,150	0,154	0,159	0,165	0,154	0,164	0,155	0,154	0,143	0,155	0,007

Table S2. Trace element compositions of magnetite and final growth medium for culture of MV-1 (this study) and AMB-1 and abiotic magnetite (Amor et al., 2015). Concentrations in magnetite correspond to average values measured for all our experiments (this study, Amor et al., 2015). Element concentration in residual solution was calculated from concentration in magnetite and average Di values determined for all experiments.

Element	Concentration in magnetite (ppm)						Concentration in residual solution (ppm)					
	MV-1	± 1SD	AMB-1	± 1SD	Abiotic	± 1SD	MV-1	± 1SD	AMB-1	± 1SD	Abiotic	± 1SD
Ag	1,8E+00	7,2E-01	3,1E+01	4,2E+01	7,8E+01	1,4E+00	2,2E-04	2,5E-06	1,2E-02		7,9E-06	2,5E-06
Al	6,0E+02	2,3E+02	4,8E+01	6,7E+01	2,2E+02	1,3E+01	7,4E-02	4,0E-03	1,5E+00	5,2E-01	1,2E+00	2,7E-02
As	2,2E+00	9,0E-01	1,3E+01	5,9E+00	1,7E+01	8,2E+00	4,7E-04	4,5E-05	7,8E-01	2,1E-01	1,9E-02	3,8E-04
B	1,2E+01	2,3E+00	4,4E+00	4,5E+00	2,4E+02	2,9E+02	3,2E-02	4,5E-04	5,3E+00	5,6E+00	9,8E-01	7,7E-02
Ba	2,8E+00	2,6E-01	1,4E+01	1,6E+01	1,3E+02	7,0E+00	2,5E-03	2,9E-05	5,1E-01	3,1E-01	8,0E-05	5,3E-05
Be							1,8E-04	1,4E-04				
Ca	5,6E+03	3,2E+03	7,1E+01	2,4E+01	2,7E+00	4,2E-02	1,5E+02	1,1E+00	6,3E+00	3,2E+00	4,1E-03	
Cd	3,0E-01	4,0E-02	1,1E+00	5,3E-01	7,0E+01	1,3E+00	1,2E-03	2,6E-05	3,9E-01	4,7E-01	2,1E-03	9,4E-05
Ce	6,9E-01	8,8E-02	5,2E+00	5,2E+00	8,1E+01	3,5E-01	3,0E-05	5,8E-07	4,9E-01	4,5E-01	1,7E-04	2,0E-04
Co	3,5E+01	7,4E-01	5,8E+01	6,5E+01	1,1E+02	1,8E+00	1,3E-01	6,9E-04	1,5E+00	1,2E+00	2,5E-05	1,1E-05
Cr	1,1E+02	2,6E+00	3,3E+02	2,7E+02	1,9E+02	4,8E-01	1,6E-02	7,0E-05	9,4E-01	1,2E-01	2,9E-04	1,9E-05
Cs			1,6E-01	4,3E-03	8,7E+00	1,3E-01	1,0E-04	7,4E-06	7,5E-01	3,2E-01	5,8E-01	6,6E-04
Cu	1,4E+02	4,4E+00	5,9E+00	4,1E+00	1,4E+02	9,9E+00	8,5E-03	2,8E-05	4,4E-01	5,0E-01	4,2E-04	1,8E-04
Dy	1,2E-02	1,0E-02					7,0E-06	3,0E-06				
Er	1,6E-02	1,0E-02					4,5E-06	1,7E-06				
Fe	3,2E+06	1,3E+05					4,5E+00	1,7E-02				
Ga	1,6E-01	1,4E-01	1,9E+01	2,7E+01	3,3E+01	1,5E+00	1,3E-04	3,2E-05	7,4E-01	2,9E-01	3,9E-01	2,8E-04
Gd	1,6E-01	4,9E-02					3,0E-05	8,9E-07				
Ge	1,5E+02	5,0E+02					9,8E-04	2,9E-03				
Hf	3,6E-02	2,5E-02					2,9E-05	1,9E-06				
Ho	6,3E-03	3,1E-03					1,2E-06	2,1E-07				
K	2,1E+02	8,8E+01	3,9E+01	3,4E+01	1,4E+02	5,2E+00	1,9E+02	5,1E+00	1,6E+03	9,3E+02	2,4E+00	1,2E-01
La	2,9E-01	2,8E-02	4,2E+00	5,0E+00	8,3E+01	1,2E+00	2,7E-05	5,9E-07	4,8E-01	4,6E-01	2,1E-05	9,4E-07
Li	1,4E-01	4,0E-02	5,3E-01	3,3E-01	7,2E+01	2,3E+00	8,3E-04	6,1E-05	8,6E-01	4,0E-01	2,3E-01	3,4E-03
Lu	1,8E-03	1,2E-03					6,7E-07	1,7E-07				
Mg	3,4E+02	6,3E+01	1,0E+02	7,2E+01	2,4E+02	7,7E+00	4,4E+02	3,7E+00	5,3E+01	5,4E+01	6,9E-03	
Mn	5,5E+01	7,6E+00	7,6E+02	8,9E+02	1,9E+03	2,6E+01	9,1E-01	5,6E-03	9,3E+00	1,2E+01	4,1E-04	1,9E-05
Mo	5,4E+00	3,4E-01	6,1E+00	1,1E+00	9,6E-01	3,1E-01	9,2E-01	3,7E-03	5,3E+00	6,8E+00	6,6E-01	6,8E-03
Na	6,5E+03	5,5E+03					8,4E+03	6,5E+01				
Nb	5,0E-01	2,7E-01					3,2E-05	1,2E-05				
Nd	1,8E+00	2,1E-01					1,5E-04	1,8E-06				
Ni	1,9E+02	3,6E+02	4,7E+01	2,6E+01	1,3E+02	7,4E-01	1,5E-01	2,8E-01	1,1E+00	2,7E-01	3,0E-04	1,6E-04
P	2,6E+03	3,8E+02					2,6E+01	7,7E-01				
Pb	3,3E+00	9,6E-02	2,8E+01	6,5E+00	7,6E+01	1,3E-01	3,7E-03	1,9E-05	6,2E-01	5,8E-01	3,8E-05	1,2E-05
Pr	5,5E-01	1,5E-01					4,2E-05	4,1E-07				
Rb	1,5E-01	1,8E-02	1,7E-01	3,0E-02	3,4E+00	1,7E-01	1,3E-02	2,2E-05	8,7E-01	8,7E-02	6,8E-01	1,7E-02
Sb	6,3E-01	7,3E-02	1,5E+01	2,0E+01	2,3E+01	5,5E-01	1,6E-04	5,0E-06	6,5E-01	4,5E-01	5,9E-04	6,6E-05
Se	4,0E-01	2,5E-01	9,1E+00	2,4E+00	4,3E+00	7,6E-01	2,5E-04	4,0E-05	5,9E-01	5,7E-01	4,0E-01	4,1E-03
Si	2,3E+03	1,3E+03					1,1E+00	2,1E-01				
Sm	6,7E-02	5,9E-02					1,0E-05	4,0E-06				
Sn	3,2E-02	3,0E-03	1,1E+02	5,3E+01	4,2E-01	0,0E+00	1,1E-04	4,5E-06	5,5E-01	2,7E-01	1,8E-01	7,3E-03
Sr	8,6E+00	1,5E-01	4,3E-01	2,1E-02	1,5E+02	2,1E+00	2,2E-02	1,0E-04	5,4E-01	9,3E-02	2,6E-04	9,4E-06
Tb	2,2E-03	1,0E-03					1,2E-06	1,4E-07				
Th	7,2E-01	1,6E+00	3,3E+01	4,6E+01	8,4E+01	7,2E-01	4,0E-05	8,1E-05	3,5E-01	4,4E-01	1,0E-05	3,5E-06
Ti	2,6E+01	1,6E+01	1,1E+02	1,2E+02	7,5E+01	1,4E+01	5,1E-03	1,1E-03	5,3E-01	9,1E-02	4,1E-04	2,7E-04
Tl	2,3E-01	6,2E-02	4,5E+00	5,3E+00	7,2E+01	1,8E+00	2,4E-05	1,1E-06	5,9E-01	2,3E-01	1,5E-04	3,5E-05
U	5,5E-01	5,6E-02	1,5E+01	2,1E+01	8,5E+01	6,5E-01	2,2E-05	8,3E-07	5,1E-01	2,6E-01	1,7E-03	2,8E-04
V	5,2E-01	1,7E-01	1,7E+00		4,2E+01	5,5E+00	2,0E-03	3,4E-05	5,1E-01	1,7E-02	4,6E-03	9,4E-05
Y	4,5E-02	1,2E-02	2,1E+00	2,7E+00	8,7E+01	2,0E+00	1,6E-05	2,3E-06	5,4E-01	4,7E-01	2,5E-05	0,0E+00
Yb	1,0E-02	1,7E-02					2,9E-06	2,9E-06				
Zn	1,4E+02	3,3E+00	6,2E+01	2,3E+01	1,4E+02	1,8E+01	2,3E-01	9,9E-04	1,5E+00	4,8E-02	5,2E-03	5,7E-03
Zr	2,5E+00	2,1E-01					1,2E-03	7,4E-06				

Table S3. Average distribution coefficients (Di) between magnetite and final growth medium for culture of MV-1. The cultures were performed under various Fe sources (Fe(II)-ascorbate and Fe(III)-citrate) at initial concentrations of 50, 100 and 150 uM.

Element	Fe(III)-citrate						Fe(II)-ascorbate					
	50 µM		100 µM		150 µM		50 µM		100 µM		150 µM	
	Average	± 1SD	Average	± 1SD	Average	± 1SD	Average	± 1SD	Average	± 1SD	Average	± 1SD
Li							1,62E+02	1,09E+02				
B	2,94E+02	3,07E+01	1,76E+02	1,25E+01	1,22E+03	2,01E+02						
Na	2,23E-01	8,94E-02	1,80E-01	6,63E-03	6,08E+00	4,92E+00	2,83E-01	1,22E-02	2,35E-01	1,85E-02	6,28E-02	1,61E-02
Mg	5,73E-02	4,42E-03	3,61E-01	9,66E-03	2,46E+00	3,81E-01	4,78E-01	1,93E-02	1,08E-01	2,77E-02	8,45E-02	2,11E-03
Al	4,19E+03	1,80E+02	7,02E+03	8,45E+02	3,14E+04	1,12E+04	4,62E+03	1,24E+03	2,63E+03	4,59E+02	2,23E+03	1,55E+02
Si	1,68E+03	1,93E+02	8,31E+01	2,04E+01			9,23E+02	2,24E+02	6,35E+03	2,57E+03		
P	1,39E+02	1,47E+01	1,28E+02	1,23E+01			7,59E+01	5,43E+00	1,11E+02	1,08E+01	1,04E+02	9,98E+00
K	1,10E+00	3,43E-01	4,88E-01	1,07E-01			8,38E-01	2,77E-01	2,47E+00	9,48E-01	6,39E-01	5,02E-01
Ca	2,28E+00	6,35E-01	1,25E+02	2,73E+00	8,85E+01	6,33E+01	2,38E+00	1,40E-01	4,59E-01	6,80E-02	6,07E-02	3,44E-03
Ti	5,92E+03	2,72E+03	2,29E+03	1,27E+03			6,04E+03	1,95E+03	4,76E+03	8,16E+02	7,11E+03	2,60E+03
V	4,02E+02	1,82E+02	1,46E+02	3,09E+01			3,05E+02	4,84E+01	2,36E+02	1,74E+01	2,73E+02	4,38E+01
Cr	9,97E+03	2,07E+02	4,09E+03	4,81E+01	1,07E+04	8,74E+01	8,65E+03	2,65E+02	6,73E+03	8,71E+01	8,20E+03	1,33E+02
Mn	8,49E+01	2,91E+00	8,16E+01	1,96E+00	6,33E+01	3,24E+01	5,70E+01	1,74E+00	6,10E+01	2,01E+00	5,79E+01	1,15E+00
Fe	3,25E+05	3,07E+03	4,13E+05	3,62E+03	7,27E+04	7,43E+02	3,36E+06	1,47E+05	2,22E+05	5,50E+03	8,01E+04	3,44E+02
Co	2,61E+02	5,25E+00	3,33E+02	4,05E+00	3,97E+02	2,28E+00	2,00E+02	6,38E+00	2,64E+02	5,94E+00	3,22E+02	2,37E+00
Ni	2,34E+03	9,02E+01	9,46E+02	2,47E+01	4,13E+03	6,43E+01	9,74E+01	7,11E+00	2,99E+02	8,81E+01	9,10E+02	2,26E+02
Cu	2,18E+03	5,45E+01	3,69E+04	6,66E+02	4,27E+04	1,36E+03	1,72E+03	5,12E+01	3,58E+03	1,24E+02	1,08E+04	8,31E+01
Zn	4,29E+02	1,74E+01	9,03E+02	9,99E+00	8,64E+02	8,40E+00	3,71E+02	1,04E+01	6,28E+02	1,80E+01	6,52E+02	7,60E+00
Ga	3,50E+03	1,32E+03	1,37E+03	1,11E+03			8,54E+02	2,96E+02	1,43E+03	1,20E+03	6,81E+02	3,49E+02
Ge	2,68E+05	6,33E+04	5,62E+05	1,77E+05	9,52E+04	2,13E+04					3,66E+04	1,21E+04
As							3,63E+03	8,50E+02	5,73E+03	2,08E+03		
Se			2,11E+03	1,14E+03							1,13E+03	3,67E+02
Rb	2,46E+01	3,63E+00	8,17E+00	6,28E-01	7,70E+01	4,33E+01	4,26E+00	2,06E+00	3,96E+00	1,51E+00		
Sr	3,46E+01	1,81E+00	2,89E+02	3,84E+00	1,94E+03	1,56E+01	2,48E+01	1,29E+00	1,53E+01	4,62E-01	5,03E+01	3,34E+00
Y	2,79E+01	2,61E+01	4,63E+03	3,78E+02			1,07E+03	3,98E+02	9,08E+02	3,71E+02	8,03E+02	2,93E+02
Zr	3,04E+03	4,12E+02	2,21E+03	9,59E+01	2,54E+03	7,54E+01	6,54E+02	1,00E+02	1,83E+03	1,30E+02	3,81E+03	2,16E+02
Nb			1,26E+04	3,11E+03			2,67E+04	2,75E+03	1,02E+04	6,87E+02		
Mo	1,11E+01	4,54E-01	7,33E+00	1,99E-01			5,48E+00	2,54E-01	6,66E+00	2,18E-01	2,90E+00	6,39E-01
Ag			1,03E+04	3,18E+03								
Cd	7,85E+01	6,24E+01	5,63E+01	1,82E+01			3,05E+01	1,36E+01			1,32E+03	3,11E+01
Sn											2,78E+02	1,50E+01
Sb	5,99E+03	7,53E+02	1,26E+03	2,00E+02			1,57E+04	8,16E+02	1,47E+03	1,49E+02	1,31E+03	2,10E+02
Cs												
Ba	1,19E+03	1,71E+02	1,08E+03	6,49E+01			6,37E+02	7,65E+01	2,01E+03	1,28E+02	1,62E+03	1,27E+02
La	7,32E+03	5,11E+02	2,41E+03	8,75E+01	1,14E+04	3,81E+02					1,44E+04	1,28E+03
Ce	2,17E+04	4,14E+03	6,10E+03	2,11E+02	2,17E+04	7,83E+02					2,86E+04	3,56E+03
Pr	6,56E+04	1,75E+04	5,98E+01	1,25E+01	2,88E+03	1,19E+03	3,68E+03	1,60E+03	3,17E+03	5,23E+02	6,89E+01	3,08E+00
Nd	8,51E+04	7,24E+03	7,61E+01	1,47E+01	2,70E+03	3,71E+02	2,21E+03	6,25E+02	2,00E+03	5,71E+02	3,06E+02	2,49E+02
Sm							9,24E+03	5,27E+03	3,72E+03	1,06E+03		
Gd	3,18E+04	7,33E+03	1,08E+02	5,26E+01			2,88E+03	1,31E+03			1,11E+02	2,52E+01
Tb			8,23E+02	6,10E+02			3,75E+03	1,02E+03	1,92E+03	4,36E+02		
Dy			1,65E+03	7,71E+02								
Ho	1,07E+03	1,00E+03	1,48E+03	4,32E+02			9,25E+03	2,96E+03				
Er			1,87E+03	7,45E+02			7,57E+03	1,58E+03	3,13E+03	1,16E+03	1,91E+03	5,69E+02
Yb									1,62E+03	1,46E+03		
Lu									2,68E+03	1,03E+03		
Hf	4,61E+02	1,36E+02							9,47E+02	4,89E+02	1,97E+03	1,60E+03
Tl					9,89E+03	2,19E+03						
Pb	1,02E+03	5,02E+01	2,80E+03	5,60E+01	6,53E+02	8,43E+00	2,54E+02	9,08E+00	1,71E+02	4,65E+00	9,00E+02	9,18E+00
Th					1,78E+04	4,69E+03						
U	3,45E+04	1,26E+03	1,83E+04	9,90E+02	2,02E+04	1,87E+03	3,29E+04	1,99E+03	3,51E+04	2,02E+03	2,26E+04	1,63E+03